Machine Learning Assignment :- 8

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Apply appropriate ML algorithm on a dataset collected in a cosmetics shop showing details of customers to predict customer response for special offer. Create confusion matrix based on above data and find

1). Accuracy

2). Precision

3). Recall

4). F-1 score

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix
```

```
data = pd.DataFrame({
    "CustomerID": [1,2,3,4,5,6,7,8,9,10,
                   11,12,13,14,15,16,17,18,19,20],
    "Age": [22,35,26,45,30,50,28,42,24,38,
            29,41,33,27,36,48,23,39,31,44],
    "Gender": [0,1,0,1,0,1,0,1,0,1,
               0,1,0,0,1,1,0,1,0,1],
    "Income": [25000,40000,32000,60000,45000,70000,38000,52000,28000,48000,
               35000,58000,42000,30000,46000,72000,26000,51000,39000,65000],
    "SpendingScore": [75,40,60,30,55,20,70,35,80,45,
                      65,25,50,85,43,15,78,38,58,33],
    "PreviousPurchases": [5,2,4,1,3,1,6,2,7,3,
                          4,1,5,8,2,1,6,2,4,1],
    "Membership": [1,0,1,0,1,0,1,0,1,0,
                   1,0,1,1,0,0,1,0,1,0],
    "Response": [1,0,1,0,1,0,1,0,1,0,
                 1,0,1,1,0,0,1,0,1,0]
})

print("Dataset:\n")
print(data)

X = data.drop(["CustomerID", "Response"], axis=1)
y = data["Response"]

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Logistic Regression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Confusion Matrix
cm = confusion_matrix(y_test, y_pred)
```

```
print("\nConfusion Matrix:")
print(cm)

TN = cm[0][0]
FP = cm[0][1]
FN = cm[1][0]
TP = cm[1][1]

accuracy = (TP + TN) / (TP + TN + FP + FN)
precision = TP / (TP + FP) if (TP + FP) != 0 else 0
recall = TP / (TP + FN) if (TP + FN) != 0 else 0
f1_score = (2 * precision * recall) / (precision + recall) if (precision + recall) != 0 else 0

print("\nEvaluation Metrics:")
print("Accuracy  :", round(accuracy,4))
print("Precision :", round(precision,4))
print("Recall    :", round(recall,4))
print("F1 Score  :", round(f1_score,4))
```

```
Dataset:

    CustomerID  Age  Gender  Income  SpendingScore  PreviousPurchases  \
0            1   22       0   25000             75                  5
1            2   35       1   40000             40                  2
2            3   26       0   32000             60                  4
3            4   45       1   60000             30                  1
4            5   30       0   45000             55                  3
5            6   50       1   70000             20                  1
6            7   28       0   38000             70                  6
7            8   42       1   52000             35                  2
8            9   24       0   28000             80                  7
9           10   38       1   48000             45                  3
10          11   29       0   35000             65                  4
11          12   41       1   58000             25                  1
12          13   33       0   42000             50                  5
13          14   27       0   30000             85                  8
14          15   36       1   46000             43                  2
15          16   48       1   72000             15                  1
16          17   23       0   26000             78                  6
17          18   39       1   51000             38                  2
18          19   31       0   39000             58                  4
19          20   44       1   65000             33                  1

    Membership  Response
0            1         1
1            0         0
2            1         1
3            0         0
4            1         1
5            0         0
6            1         1
7            0         0
8            1         1
9            0         0
10           1         1
11           0         0
12           1         1
13           1         1
14           0         0
15           0         0
16           1         1
17           0         0
18           1         1
19           0         0

Confusion Matrix:
[[2 0]
 [0 4]]

Evaluation Metrics:
Accuracy  : 1.0
Precision : 1.0
Recall    : 1.0
F1 Score  : 1.0
```